LAKSHMANAN T

Software Engineer — Backend · AI · Full-Stack | B.E. Computer Science (2024–2028)

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lakshmanan272.github.io/MY-PORTFOLIO-LAKSHMANAN | github.com/lakshmanan272 | linkedin.com/in/lakshmanan-t-747981372 | leetcode.com/u/Lakshmanan-T

SUMMARY

Computer Science student who builds backend systems that stay correct under load (Java/Spring Boot, a Redis clone in C) and AI products people actually use. Built a placement portal my class has run on for 3 months. Inventor on 2 published Indian patent applications, IEEE-published author, and 329 LeetCode problems solved (51 hard).

TECHNICAL SKILLS

Languages: Java, C, Python, TypeScript, JavaScript, SQL
Backend: Spring Boot 3, Spring Security (JWT), Spring Data JPA, REST APIs, Node.js, Express, Flask
Frontend: React, Next.js, Tailwind CSS, Three.js · **Data & DevOps:** PostgreSQL, MongoDB, Prisma, Docker, Git, Vercel, Render
AI & CS fundamentals: ElevenLabs Agents, MediaPipe, Vosk, NetworkX, LSTM/GNN · Data Structures & Algorithms, OOP, Concurrency, DBMS

EDUCATION

B.E. Computer Science and Engineering — V.S.B. College of Engineering Technical Campus, Coimbatore — 2024 – 2028 · **CGPA 8.6 / 10**

PROJECTS

BookMyEvent — Oversell-Safe Ticket Booking — bookmyevent-fawn.vercel.app · github.com/lakshmanan272/bookmyevent

- Booking path guarded by a pessimistic row lock (*SELECT ... FOR UPDATE*); an integration test races 30 threads for 5 seats with zero oversells.
- Stateless JWT auth with BCrypt and USER/ADMIN roles; Caffeine-cached event catalog, rate limiter and Spring Actuator.
- Pluggable distributed lock (DB / Redis / MiniRedis); multi-stage Docker build, docker-compose and a k6 load test; deployed on Vercel + Render.

Tech: Java 17, Spring Boot 3, Spring Security, JPA, PostgreSQL, Next.js 14, TypeScript, Docker

MiniRedis — Redis Clone in C — github.com/lakshmanan272/miniredis

- Implements the real RESP protocol over TCP (redis-cli compatible): SET [NX] [EX|PX], GET, DEL, EXPIRE, TTL, INCR/DECR, KEYS, INFO.
- TTL expiry like Redis (lazy + background sweeper); SET NX PX lock with token-checked release, verified by a 60-thread race (exactly one winner).
- Hand-written FNV-1a hash table, thread-per-connection server and HTTP gateway; serves as BookMyEvent's lock provider. Zero external dependencies.

Tech: C11, POSIX sockets, pthreads, Docker (Alpine)

Placement Portal — in production for 3 months — Next.js 14 · TypeScript · Prisma · PostgreSQL (Neon)

- Used every day by my class of 63 students, the placement cell and the class teacher to run hackathon and internship registrations.
- Race-safe first-come team slots via database transactions; one reusable student profile; role-based views for 3 user types; mobile-first UI.

Kadai AI — Voice Supermarket Agent — Next.js · TypeScript · ElevenLabs Agents

- Customers order by voice in Tamil, Tanglish or English; the agent searches products, builds a live cart, applies offers and issues a QR order ticket.

Jarvis — Offline Desktop Voice Agent — Python · Vosk · OpenWakeWord · Win32 · github.com/lakshmanan272/agent-jarvis

- Offline speech recognition and wake word; ~21 µs intent routing; clicks, types, opens apps and controls the system in under 400 ms.

INTERNSHIPS & TRAINING

Full Stack Web Development Intern — CODTECH IT Solutions, Hyderabad (remote) — Jun – Jul 2025
Full Stack Web Development Intern — Pantech Prolabs India Pvt Ltd — Jun 2025
30-Day MasterClass in Full Stack Development — NoviTech R&D, Coimbatore — Sep – Oct 2024

PATENTS & PUBLICATIONS

- **Patent (India, pending):** Electrifying the Future by Intersecting Windmills and Tesla Coils — App. 202441089752 A, filed 19 Nov 2024
- **Patent (India, pending):** GaiaNet: A Quantum-AI Powered Survival Shield for the Future of Humanity — App. 202541088719 A, filed 17 Sep 2025
- **IEEE, ICCPCT 2026:** Detection of REM Sleep Behavior Disorder Using LSTM and GNN Model — DOI 10.1109/ICCPCT70290.2026.11654757
- **ICSAIC 2026:** Cognitive Phantom Networks — AI hallucination as a self-replicating attack surface (paper presented)

CERTIFICATIONS & ACHIEVEMENTS

- **NPTEL Programming in Java — Elite + Gold (92%)**, IIT Kharagpur · NPTEL Cloud Computing · Infosys Springboard Angular · Anthropic AI Fluency · CSC Advanced Diploma in Python (Grade A)
- Shortlisted in the Smart India Hackathon 2025 college internal round · Hackathons: Shaastra 2026 (IIT Madras), Pragyan 2026 (NIT Trichy), DEVSPARK'25 (KPR IET), CRYPTERA 2026 (CIT), Adobe University Hackathon